Graphical Abstract

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Highlights

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Longitudinal Medical Image Imputation via Bidirectional Anatomy-Guided Diffeomorphic Deformations

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Abstract

Keywords:

1. Introduction

In medical imaging, longitudinal data plays a central role in evaluating how anatomy changes over time. Clinicians rarely interpret a scan in isolation; instead, they often assess follow-up images with respect to a baseline, using the subject's own imaging history as the reference point. This approach accounts for natural anatomical variability and enables a more personalized judgment of what constitutes as progression.

These trajectories are shaped by a range of subject-specific factors, including chronological age, pre-existing conditions, or genetic risk, that influence both the pattern and rate of change. An older subject, for example, may show substantial atrophy even in the absence of disease, while another may exhibit more abrupt degeneration due to pathological processes. In this context, the interpretation of a scan depends as much on where the subject started as on where they are now.

Yet despite its clinical value, longitudinal imaging data is often incomplete. Missed visits, dropouts, and inconsistent imaging protocols introduce sparsity across timepoints, making it difficult to reconstruct a coherent picture of disease progression. When available, complete trajectories offer the possibility of capturing gradual trends as well as detecting deviations from them—events that might otherwise be missed but could carry clinical significance. For data-driven models, such irregularity complicates the learning of

coherent temporal patterns and introduces risk of bias in downstream tasks [**?** **?** **?**].

Generative models offer a natural framework for scan imputation, and recent work in GANs and diffusion models has demonstrated strong results in image synthesis. Still, many of these methods operate directly in intensity space, generating voxel values anew. In high-dimensional medical images, this can lead to anatomically implausible results, especially when training data is limited. Some approaches synthesize 2D slices and reassemble them into volumes [**?**], at the risk of discontinuities across planes. Others model deformation trajectories explicitly [**?**], but require separate models per subject and per timepoint, limiting generalization and scalability.

Additionally, in most longitudinal scans, most anatomical regions remain unchanged between timepoints. These areas offer limited insight into disease progression. It is the evolving regions—subtle, spatially localized, and shaped by each subject's trajectory, that carry the signal of interest. From a clinical perspective, what matters is not the absolute appearance of a scan, but how it differs from where the subject began.

This suggests that modeling transformation, rather than appearance, might offer a more effective path forward. Anchoring all timepoints to a common baseline reflects how clinicians reason about progression, and provides a consistent frame of reference across individuals. In doing so, it also reduces inter-subject variability, which facilitates comparison.

This perspective is particularly relevant in conditions like Alzheimer's disease, where two subjects may show comparable reductions in tissue volume at a given follow-up, though only one may be undergoing pathological decline. This distinction has significant clinical implications, especially in early-stage intervention and patient-specific monitoring. Modeling change relative to baseline helps distinguish such cases by restoring the temporal context.

A well-established representation of anatomical change in neuroimaging is the stationary velocity field (SVF), widely used in deformable registration [**?** **?** **?**]. SVFs define smooth, invertible transformations between images, offering a compact, geometrically grounded way to encode how anatomy deforms over time. Rather than synthesizing entire images, one estimates the transformation required to reach the desired anatomical state. This shifts the modeling focus from raw appearance to subject-specific progression, which is arguably the central object of interest in longitudinal analysis.

In this paper, we introduce SynBADD (Longitudinal Medical Image Im-

putation via **Syn**thetic **B**idirectional **A**natomy-Guided **D**iffeomorphic **D**eformations), a generative framework for imputing missing 3D scans by predicting anatomically plausible deformation fields from the subject's baseline. Rather than synthesizing voxel intensities, SynBADD estimates the diffeomorphic transformation that would align the baseline to the missing timepoint. This encourages consistency with anatomical structure, and constrains generation to plausible deformation trajectories. A transformer-based model learns to reason jointly over past and future changes, capturing nonlinear and individualized progression patterns. To address the scarcity of real longitudinal data, we pre-train a variational autoencoder on a large corpus of synthetically generated SVFs, learning a compact latent space of deformation dynamics. A refinement module then enhances the predicted field to preserve fine-grained anatomical detail.

Our contributions are as follows:

- We introduce a novel trajectory-conditioned deformation-based generative model that imputes missing intermediate scans by leveraging both past and future anatomical information across subjects.
- We propose a compact latent representation of SVFs learned from synthetic data, enabling effective learning under limited data by decoupling deformation modeling from raw intensity generation.
- We design a spatial refinement module that improves the fidelity of the predicted deformation, preserving subject-specific anatomical detail.
- We achieve state-of-the-art results on the ADNI dataset, improving SSIM by 2.39% and PSNR by 10.64%, while maintaining anatomically coherent progression in synthesized scans.

2. Background and Related Works

2.1. Missing Data in Medical and Longitudinal Imaging

Missing data is a common and significant challenge in clinical imaging studies, particularly in longitudinal settings where subjects undergo repeated scans over time. Data may be missing at one or more timepoints due to various factors such as motion artifacts, missed visits, or participant dropout. This problem is especially pronounced in aging or disease-prone populations,

where frequent imaging is required to monitor disease progression, thereby increasing the likelihood of incomplete scan sequences.

Traditional approaches to handling missing data, such as list wise deletion where subjects with incomplete imaging records are excluded, can substantially reduce the effective sample size and introduce selection bias, ultimately compromising the validity and generalizability of the study findings []. Alternative imputation techniques, including last observation carried forward and linear interpolation, have been applied in certain contexts. However, these methods are primarily designed for structured, low-dimensional data and are not well suited to the spatial and contextual complexity inherent in high-dimensional modalities such as three-dimensional volumetric imaging.

In medical imaging, missing data can arise at multiple levels, including within individual scans, for example when slices are absent or corrupted, and across temporal sequences, such as when entire follow-up volumes are unavailable. While deep learning models such as conditional generative adversarial networks and convolutional neural networks have demonstrated success in intra-scan slice imputation, these approaches do not readily extend to the more complex task of reconstructing full three-dimensional scans at missing timepoints in a longitudinal imaging sequence.

2.2. Predicting Future Scans in Longitudinal Medical Imaging

Predictive modeling of future scans enables proactive assessment of anatomical and pathological progression. Early approaches in longitudinal imaging relied on parametric models such as linear mixed-effects models [] and geodesic regression [] to capture temporal trends in derived clinical or morphometric features, such as brain volume or cortical thickness. While interpretable, these methods often assume smooth, low-dimensional trajectories that may not reflect the heterogeneous and nonlinear progression patterns typical of neurodegenerative diseases. More recent deep learning approaches directly predict future scans from past imaging data, using architectures such as recurrent neural networks and convolutional encoder-decoders []. However, many of these models are trained in a unidirectional manner, relying solely on preceding scans for forward prediction, which limits their applicability when intermediate timepoints are missing.

Recent models such as ReMiND [] and NoDeR [] address this limitation by incorporating temporal context for scan imputation. ReMiND employs a diffusion model conditioned only on the immediate past and future scans, synthesizing missing images in intensity space without enforcing anatomical cor-

respondence. NODER, in contrast, learns a subject-specific trajectory in the latent space of stationary velocity fields (SVFs), enabling intermediate scan prediction through curve fitting. However, its coarse supervision may lead to oversmoothing or inaccurate estimation of anatomical changes. SynBADD adopts a distinct approach by learning deformation fields—parameterized as stationary velocity fields (SVFs)—from the baseline to the missing timepoint, while leveraging the full longitudinal context. By operating in deformation space rather than directly synthesizing intensities, SynBADD preserves subject-specific anatomy, maintains diffeomorphic consistency, and supports temporally coherent interpolation at discrete timepoints.

2.3. Generative Models for Medical Image Synthesis

Generative models are increasingly being integrated into medical image analysis workflows, supporting tasks such as modality translation, super-resolution, anomaly detection, and missing data imputation. Common generative frameworks include Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and diffusion models. VAEs [] provide latent representations that support interpolation and uncertainty modeling, but their outputs are often blurry due to the regularization imposed by their probabilistic formulation. Despite this, VAEs have been used effectively in tasks involving anatomical variation and longitudinal modeling []. GANs [], in contrast, generate sharper images through adversarial training and have been applied to tasks such as cross-modality synthesis [], slice inpainting [], and super-resolution []. However, they are prone to challenges such as mode collapse and training instability, which can limit their reliability and generalizability in clinical applications. Diffusion models [] have recently gained traction for their strong performance in generating high-resolution images. These models synthesize data by iteratively denoising random noise and are known for their stable training dynamics and flexible conditioning mechanisms. However, their applicability can be limited by slow generation times, primarily due to the computational cost of iterative reverse sampling, which becomes even more pronounced when generating large three-dimensional volumetric scans. ReMiND [] applies diffusion modeling to the task of longitudinal imputation by synthesizing intermediate MRI slices in two dimensions, which are stacked along the depth axis in groups of 12 slices to approximate a three-dimensional volume. The generation is conditioned on both the preceding and succeeding timepoint scans. However, it operates in image space and is designed to interpolate between a fixed pair of scans, rather than leverag-

ing the full longitudinal sequence for context-aware imputation. Moreover, it lacks explicit constraints to ensure anatomical coherence across the full 3D volume or over time. In longitudinal imaging, preserving subject identity and modeling smooth anatomical progression is essential—especially in neurodegenerative studies where changes are subtle and individualized. Direct voxel-level synthesis may overlook underlying structural transformations. SynBADD addresses this by shifting the generative focus from intensities to deformations. It learns to produce stationary velocity fields (SVFs) that deform a baseline scan to approximate a missing timepoint, ensuring diffeomorphic consistency and temporally structured interpolation across multiple, discretized timepoints.

2.4. Deformation-Based Approaches in Image Modeling

Deformation-based models describe anatomical variation by estimating spatial transformations between images. Classical frameworks such as Large Deformation Diffeomorphic Metric Mapping (LDDMM) [] model smooth, invertible deformations that preserve topological structure, making them well-suited for tasks like registration and morphometric analysis. More recently, neural network-based approaches like VoxelMorph [] have enabled fast, learning-based registration by training convolutional networks to predict deformation fields directly from image pairs. These methods significantly reduce inference time and support population-level learning, but are typically designed for pairwise alignment and do not naturally generalize to synthesis or imputation tasks. In this work, we extend deformation modeling from alignment to generation. Rather than predicting a deformation between two observed scans, we train a model to synthesize the deformation that maps a baseline image to a missing timepoint. This requires deformations that are both anatomically meaningful and temporally consistent. To this end, we represent transformations using stationary velocity fields (SVFs), which define time-invariant velocity fields whose exponential map yields smooth, diffeomorphic transformations. Compared to displacement fields, SVFs offer a compact, invertible, and topologically stable representation of anatomical change. Crucially, our model predicts deformations from the baseline to each target timepoint, rather than estimating them piecewise between adjacent scans. This baseline-referenced formulation maintains a consistent anatomical coordinate frame across all predictions, improving interpretability and avoiding the accumulation of errors that can arise from chaining local transformations. Unlike sequential methods, which may over-smooth pro-

gression or drift in the presence of irregular sampling, our approach enforces a single, well-defined transformation per timepoint. This results in anatomically coherent and temporally aligned interpolation, making it well-suited for modeling longitudinal trajectories in neuroimaging.

2.5. Alzheimer's Disease and Longitudinal Imaging

Alzheimer's disease (AD) is a progressive neurodegenerative condition characterized by gradual, region-specific brain atrophy—most notably affecting the hippocampus, entorhinal cortex, along with certain cortical structures []. These anatomical changes unfold over years and vary considerably across individuals, making longitudinal imaging a critical tool for studying disease progression, stratifying risk, and informing clinical decision-making. The Alzheimer's Disease Neuroimaging Initiative (ADNI) has played a central role in enabling such research by providing a large-scale, multi-site repository of structural MRI scans acquired at multiple timepoints along the disease spectrum. However, as with many longitudinal studies, ADNI is affected by missing data due to participant dropout, missed visits, and irregular scan intervals. The presence of missing data complicates temporal modeling and can introduce bias when models require complete sequences have to rely on naive imputation techniques. In this context, models capable of anatomically consistent imputation are particularly valuable. SynBADD has been designed with this setting in mind, reconstructing plausible intermediate scans by predicting subject-specific deformation fields based on baseline anatomy and available surrounding timepoints. Rather than generating missing scans directly in intensity space, SynBADD models anatomical progression through predicted deformation fields. By learning stationary velocity fields (SVFs) that map the baseline anatomy to a missing timepoint, it captures subject-specific patterns of neurodegeneration in a temporally consistent and anatomically meaningful way. This approach allows SynBADD to impute intermediate scans even when only partial longitudinal data are available, reducing reliance on fully sampled trajectories. The deformation-based formulation preserves structural continuity and supports robust modeling of disease progression, enabling a more reliable use of incomplete longitudinal imaging data sets in downstream clinical analysis.

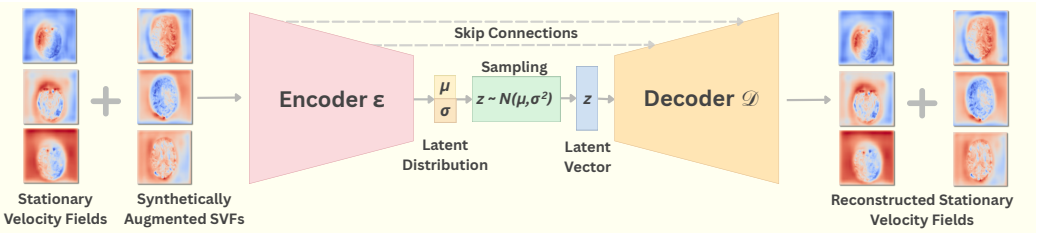


Figure 1: Learning compact latent representation of SVFs with synthetic augmentation for improved deformation modelling.

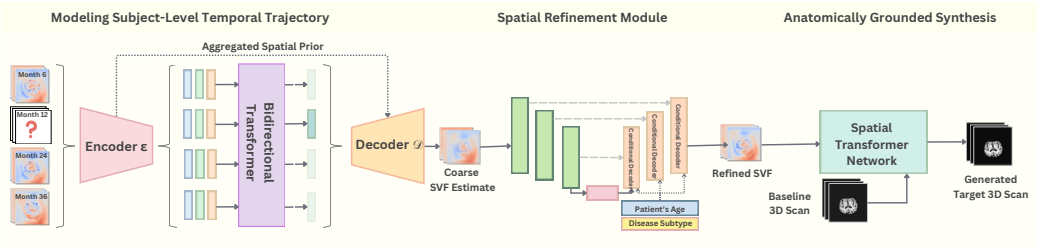


Figure 2: An overview of our proposed framework, SynBADD: learning compact SVF representations, modeling subject-specific temporal trajectories, refining coarse SVF estimates and synthesizing missing scans by warping the baseline image.

3. Method

We address the problem of synthesizing a missing brain MRI scan at an intermediate timepoint in a subject's longitudinal imaging sequence by modelling anatomical deformations over time. Consider a subject with scans acquired at discrete timepoints $\{T_0, T_1, \ldots, T_n\}$, where T_0 represents the baseline or initial screening scan obtained upon the subject's enrollment in the study and T_i represents the scan at i^{th} timepoint. Let $T_p \in \{T_1, \ldots, T_n\}$ denote the timepoint at which the scan is missing. Each available scan I_{T_i} is registered to the baseline I_0 using symmetric diffeomorphic registration via ANTs toolkit []. This produces a stationary velocity field (SVF), denoted $\phi_{0\to T_i}$, which encodes the deformation from the baseline to the timepoint T_i. These SVFs are used to learn latent representations of anatomical changes in the subject over the given timepoint, which are then used to predict the deformation for the missing timepoint.

We aim to estimate the deformation $\hat{\phi}_{0\to T_p}$ which corresponds to the SVF from the baseline scan to the missing timepoint T_p. The missing image $\hat{I}_{T_p}$ is

synthesized by warping the baseline scan I_0 as per this predicted deformation. We approach this problem with three key assumptions. First, anatomical changes progress smoothly over time, even though the trajectory may not be strictly linear and may vary across subjects. Second, each deformation can be represented in a low-dimensional latent space that preserves both global structure and local anatomical detail. Third, the neighbouring timepoints of a subject can provide a useful context to infer the deformation at a missing timepoint. To address this, we propose a three-stage framework. We first learn a compact latent representations of the SVFs using a variational Auto-encoder. Next, we use a target-aware temporal transformer to infer the latent deformation at the missing timepoint by attending to relevant information across the available timepoints. Finally, the predicted deformation is refined and applied to the baseline scan to synthesize the missing image in an anatomically consistent manner.

3.1. Latent Representation Learning of SVFs

We learn a compact latent representation for each SVF using a Variational Auto-Encoder (VAE) with skip-connections. Each deformation field $\phi_{0\to T_i}$ is independently encoded into a latent vector $z_i \in \mathbb{R}^d$ using a convolutional encoder $\mathcal{E}$. In parallel, the encoder also extracts a hierarchy of skip features s_i that preserve spatial detail at multiple resolutions:

$$z_i, s_i = \mathcal{E}(\phi_{0\to T_i})$$

The decoder $\mathcal{D}$ reconstructs the original SVF from the latent code and the corresponding skip features:

$$\hat{\phi}_{0\to T_i} = \mathcal{D}(z_i, s_i)$$

The VAE is trained to minimize a loss that includes a reconstruction term between the input and predicted SVF, and a Kullback-Leibler divergence term that regularizes the latent space and EPE and Anglular error loss. This formulation encourages the model to learn smooth, compact embeddings while preserving anatomical detail.

Most subjects only have a few longitudinal scans available which limits the number of training samples. We supplement this by introducing a synthetic SVF augmentation strategy to improve the robsutness of the learned representations. We generate additional augmentations by applying random

affine and smooth non-linear transformations to the input fields. These augmentations preserve diffeomorphic properties and introduce variability that helps the model generalize to unseen deformation patterns.

At inference time, the skip features $\{s_j\}_{j \neq p}$ from available timepoints are averaged to create a subject-specific spatial prior:

$$\bar{s} = \frac{1}{n-1} \sum_{j \neq p} s_j$$

This aggregated skip representation is used during decoding of the predicted latent code $\hat{z}_p$ to help preserve anatomical consistency when reconstructing the missing deformation.

3.2. Trajectory Inference via Target-Aware Temporal Transformer

Given a sequence of latent representations $\{z_1, \ldots, z_n\}$ corresponding to a subject's longitudinal SVFs, our objective is to estimate the latent code $\hat{z}_p$ for a missing timepoint T_p. To address this, we propose a target-aware temporal transformer that selectively aggregates information from available timepoints, conditioned on the index of the missing timepoint. Specifically, let t_p denote the index of the missing timepoint. We compute a query embedding $\mathbf{q}_{t_p} \in \mathbb{R}^d$ using a learnable linear projection $L : \mathbb{Z} \to \mathbb{R}^d$, such that $\mathbf{q}_{t_p} = L(t_p)$. This embedding is used to guide the attention mechanism within the transformer, allowing the model to focus on temporally relevant latent vectors from the sequence for reconstructing the missing scan.

The input to the transformer consists of latent vectors $\{z_1, \ldots, z_n\}$, each representing an observed timepoint. To incorporate temporal order, we add a learnable positional embedding e_i^{pos}:

$$x_i = z_i + e_i^{\text{pos}}$$

This augmented sequence $\{x_1, \ldots, x_n\}$ is then passed through a transformer encoder, which captures contextual relationships and temporal dynamics across the observed timepoints:

$$H = \text{Transformer}(\{x_1 \ldots x_n\}) \in \mathbb{R}^{T \times D}$$

To guide the model's attention toward more informative timepoints, we apply soft-selection via gated attention. For each token H_i, we compute an

attention score $a_i = \mathrm{MLP}_{\mathrm{attn}}(H_i)$ and a gating value $g_i = \sigma(\mathrm{MLP}_{\mathrm{gate}}(H_i))$. These are combined to produce a soft attention weight:

$$\alpha_i = \frac{\exp(a_i \cdot g_i)}{\sum_j \exp(a_j \cdot g_j)}$$

The resulting gated sequence is obtained by reweighting each token:

$$\tilde{H}_i = \alpha_i \cdot H_i$$

To extract a representation specific to the missing timepoint, we use the previously obtained learned embedding $\mathbf{q}_{t_p} = L(t_p)$ associated with the target timepoint's index t_p in the longitudinal sequence. This target-aware query attends to the gated sequence using scaled dot-product attention:

$$\beta_i = \frac{\exp(q^\top \tilde{H}_i)}{\sum_j \exp(q^\top \tilde{H}_j)}$$

and produces the pooled representation as a weighted combination of temporal features:

$$\hat{z}_p = \sum_i \beta_i \tilde{H}_i$$

By explicitly conditioning both the attention and pooling mechanisms on the target timepoint, we hypothesize the model is able to selectively extract temporal context relevant to the missing deformation.

3.3. Image Synthesis with Anatomy-Preserving Warping

The predicted latent code $\hat{z}_p$ for the missing deformation is decoded using the pretrained decoder $\mathcal{D}$ introduced earlier to reconstruct the SVF. To preserve spatial detail of the subject's morphology in the SVF's representation, we also pass the aggregated skip features $\bar{s}$ computed from the available timepoints to the decoder:

$$\tilde{\phi}_{0 \to T_p} = \mathcal{D}(\hat{z}_p, \bar{s})$$

This results in a coarse estimate of the SVF for the missing timepoint. To refine this estimate and improve anatomical plausibility, we introduce a U-Net–based deformation generator $\mathcal{G}$ that takes the coarse SVF and additional clinical metadata as input. Specifically, we condition the network on

the subject's age, diagnostic label and the target timepoint index using dynamic affine feature transform (DAFT), which injects these variables into the feature maps at multiple layers:

$$\hat{\phi}_{0\to T_p} = \mathcal{G}(\tilde{\phi}_{0\to T_p}, \text{age}, \text{label}, t_p)$$

The final deformation field $\hat{\phi}_{0\to T_p}$ is then used to synthesize the missing image by warping the baseline scan I_0 using a spatial transformer network (STN):

$$\hat{I}_{T_p} = \text{STN}(I_0, \hat{\phi}_{0\to T_p})$$

All synthesis operations are performed at the original resolution to ensure fidelity to fine-grained anatomical structures. This completes the pipeline by producing a fully reconstructed image at the target timepoint.

3.4. Training Objectives

SynBADD is trained in two stages, each with a dedicated set of loss functions. The first stage trains the variational autoencoder to learn compact latent representations of SVFs. The second stage trains the temporal transformer and spatial refinement module end-to-end for missing scan synthesis.

Stage 1: VAE Pretraining

The VAE is trained to reconstruct input SVFs while learning a regularised latent space. The objective combines four terms:

$$\mathcal{L}_{\text{VAE}} = \lambda_{\text{recon}}\mathcal{L}_{\text{recon}} + \lambda_{\text{KL}}\mathcal{L}_{\text{KL}} + \lambda_{\text{EPE}}\mathcal{L}_{\text{EPE}} + \lambda_{\text{ang}}\mathcal{L}_{\text{ang}} \tag{1}$$

The reconstruction loss $\mathcal{L}_{\text{recon}}$ is the mean squared error between the input SVF $\phi_{0\to T_i}$ and the reconstructed field $\hat{\phi}_{0\to T_i}$, computed voxel-wise across all three spatial components of the velocity field. The Kullback–Leibler divergence $\mathcal{L}_{\text{KL}}$ regularises the approximate posterior $q(z_i|\phi_{0\to T_i})$ towards a standard normal prior, encouraging a smooth and compact latent space.

To encourage fidelity in both the magnitude and direction of the reconstructed velocity vectors, we include two additional terms. The end-point error $\mathcal{L}_{\text{EPE}}$ measures the average Euclidean distance between the predicted and ground-truth velocity vectors at each spatial location:

$$\mathcal{L}_{\text{EPE}} = \frac{1}{|\Omega|}\sum_{x\in\Omega} \|\hat{v}(x) - v(x)\|_2 \tag{2}$$

The angular error loss $\mathcal{L}_{\text{ang}}$ penalises directional misalignment between predicted and ground-truth vectors, computed as the average angular deviation across spatial locations:

$$\mathcal{L}_{\text{ang}} = \frac{1}{|\Omega|} \sum_{x \in \Omega} \arccos\left(\frac{\hat{v}(x) \cdot v(x)}{\|\hat{v}(x)\|_2 \, \|v(x)\|_2 + \epsilon} \right) \tag{3}$$

where ϵ is a small constant for numerical stability. Together, EPE and angular error ensure that the reconstructed SVF preserves both the scale and orientation of the original deformation field, which is important for maintaining anatomical plausibility in downstream synthesis.

Stage 2: Refinement and Synthesis

The refinement module G and the spatial transformer network are trained jointly using a composite objective that combines deformation-level and image-level losses:

$$\mathcal{L}_{\text{synth}} = \lambda_{\text{def}} \mathcal{L}_{\text{def}} + \lambda_{\text{SSIM}} \mathcal{L}_{\text{SSIM}} + \lambda_{\text{MSE}} \mathcal{L}_{\text{MSE}} + \lambda_{\text{Jac}} \mathcal{L}_{\text{Jac}} \tag{4}$$

The deformation energy loss $\mathcal{L}_{\text{def}}$, inspired by LDDMM formulations [], encourages smooth, diffeomorphic deformation fields by penalising excessive deformation energy:

$$\mathcal{L}_{\text{def}} = \int_{\Omega} \|\nabla \hat{\phi}_{0 \to T_p}(x)\|^2 \, dx \tag{5}$$

This term acts as a spatial regulariser, discouraging high-frequency deformation artifacts and promoting smoothly varying velocity fields.

The structural similarity loss $\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(\hat{I}_{T_p}, I_{T_p})$ promotes perceptual consistency between the synthesised scan $\hat{I}_{T_p}$ and the ground-truth target I_{T_p}, while the mean squared error loss $\mathcal{L}_{\text{MSE}} = \|\hat{I}_{T_p} - I_{T_p}\|^2$ enforces voxel-level intensity agreement. These two terms operate on the final warped image rather than on the deformation field itself, providing direct supervision on the synthesised output.

The Jacobian determinant regularisation $\mathcal{L}_{\text{Jac}}$ constrains the local volume change induced by the predicted deformation to remain within a biologically plausible range:

$$\mathcal{L}_{\text{Jac}} = \frac{1}{|\Omega|} \sum_{x \in \Omega} \Big(\det(J_{\hat{\phi}}(x)) - \tau \Big)^2 \tag{6}$$

where $\det(J_{\hat{\phi}}(x))$ is the Jacobian determinant of the predicted deformation at location x, and $\tau = 0.9$ is the target value. A target slightly below unity reflects the mild tissue contraction expected in a cohort that includes subjects undergoing neurodegenerative progression, where average atrophy rates in the range of 0.3–5% per year have been reported depending on clinical stage []. This regularisation discourages both excessive shrinkage and implausible tissue expansion, anchoring the predicted deformations to physiologically realistic patterns.

The loss weights are set as $\lambda_{\text{def}} = 0.5$, $\lambda_{\text{SSIM}} = 0.05$, $\lambda_{\text{MSE}} = 0.05$, and λ_{Jac} is tuned on the validation set. The higher weight on the deformation energy term reflects its importance in maintaining diffeomorphic properties, while the image-level losses provide complementary supervision on the final output.

4. Experimental Settings

4.1. Dataset

We conduct our experiments on the ADNI-1 longitudinal cohort [], which comprises T1-weighted 3D brain MRI scans acquired using 1.5T scanners at six scheduled timepoints: baseline (screening), 6 months, 12 months, 18 months, 24 months, and 36 months. The dataset includes 382 participants, of whom 148 were diagnosed with Mild Cognitive Impairment (MCI), 99 with Alzheimer's Disease (AD), and the remainder classified as cognitively normal (CN). For evaluation, we employ five-fold cross-validation with subject-level stratified splitting. This ensures that all scans from a given subject appear exclusively in a single fold, preventing data leakage across training and validation sets.

4.2. Preprocessing

All scans were preprocessed using FreeSurfer v8.0.0 []. We applied standard corrections including B1 bias field correction and N3 non-uniformity correction, followed by intensity normalization using `mri_convert`. When scan metadata did not include scaling information, a default scale factor of 1.0 was used. For scans lacking prior skull stripping, brain extraction was performed using `mri_watershed`. To account for inter-scan variability while preserving anatomical structure, each follow-up scan is coregistered to its respective baseline scan using rigid-body affine transformation via ANTs. Following registration, all scans were zero-padded to a cubic volume of $(256, 256, 256)$

voxels, then downsampled to $(128, 128, 128)$ to reduce computational overhead while preserving anatomical structure. To represent anatomical change over time, we compute stationary velocity fields (SVFs) between each follow-up scan and the baseline using VoxelMorph []. These SVFs were then used to warp the follow-up scans onto the baseline space. The warped follow-up scans, aligned to the baseline anatomy, serves as the training targets for our deformation-based generative framework.

4.3. Implementation Details

Network Architectures

The VAE encoder and decoder follow a 3D U-Net architecture [] with five convolutional layers and multi-resolution skip connections. Each convolutional block consists of $3 \times 3 \times 3$ convolutions followed by instance normalisation and leaky ReLU activations. The encoder produces a latent vector $z_i \in \mathbb{R}^d$ along with a hierarchy of skip features s_i at each resolution level. The decoder mirrors the encoder and reconstructs the SVF from the latent code and skip features.

The temporal trajectory model is a transformer encoder [] with two layers and eight attention heads, operating over a sequence of up to five latent codes corresponding to the longitudinal timepoints following the baseline. Each latent code is augmented with learnable positional embeddings encoding the relative temporal ordering and time directionality embeddings distinguishing past from future observations relative to the missing timepoint. A learnable linear projection maps the target timepoint index to a query embedding $q_{t_p} \in \mathbb{R}^d$ used for target-aware attention pooling.

The spatial refinement network is a 3D U-Net that takes the coarse decoded SVF as input and produces the refined deformation field. Clinical metadata—specifically subject age and diagnostic label—are injected at every layer of this network via dynamic affine feature-map transforms (DAFT) [], which apply learned affine transformations to the intermediate feature maps conditioned on the metadata variables. The final warping is performed using a differentiable spatial transformer network (STN) [] at the full spatial resolution of $128 \times 128 \times 128$.

Synthetic Augmentation

During VAE pretraining, we apply on-the-fly synthetic augmentations to expand the effective training set of SVFs. Augmented fields are generated through random combinations of three transformation types: random

3D rotations sampled from the SO(3) group using Euler angle parameterisation, smooth affine ramp fields that introduce spatially varying scaling and shearing, and elastic noise perturbations constrained to maintain smoothness. These augmentations are applied directly to the SVFs and are designed to preserve diffeomorphic properties, ensuring that the augmented fields remain physically plausible deformations. This strategy enriches the intra-subject deformation space and improves the model's capacity to generalise across unseen subjects and anatomical variations, which is particularly important given the limited number of real longitudinal scans per subject.

Training Procedure

All models are trained using four NVIDIA RTX 6000 GPUs (48 GB memory each). We employ five-fold cross-validation with subject-level stratified splitting, ensuring that all scans from a given subject appear in exactly one of the training, validation, or test partitions. Each fold contains approximately 275 subjects (1100 samples) for training, 30 subjects (120 samples) for validation, and 77 subjects (320 samples) for testing. Each subject contributes between three and five timepoints.

The optimiser is Adam with an initial learning rate of 5×10^{-5}. We use a ReduceOnPlateau learning rate scheduler with a patience of 5 epochs, reducing the learning rate when the validation loss plateaus. Training is terminated via early stopping with a patience of 7 epochs based on validation loss. The batch size is set to 6 for all experiments. The best-performing model checkpoint, selected based on the lowest validation loss, is used for all final evaluations on the held-out test set.

Parcellation

To ensure accurate delineation of each region of interest (ROI), anatomical parcellation is performed on the original-resolution images prior to skull stripping and preprocessing. The resulting parcellation maps are then subjected to the same preprocessing pipeline as the MRI volumes to maintain spatial correspondence. Finally, the parcellations are registered to the corresponding preprocessed scans using nearest-neighbour interpolation, preserving discrete label integrity and ensuring correct alignment between the parcellation maps and the volumetric data.

4.4. Metrics

We evaluate SynBADD along three complementary axes: image-level fidelity, region-level anatomical accuracy, and deformation-level plausibility.

All metrics are computed on the brain-masked region and averaged over the five cross-validation folds.

Image-Level Metrics

Peak signal-to-noise ratio (PSNR) quantifies voxel-level reconstruction accuracy as the ratio between the maximum possible signal intensity and the mean squared error between the synthesised and ground-truth scans. Higher values indicate greater intensity fidelity. The structural similarity index measure (SSIM) [] evaluates perceptual similarity by comparing local patterns of luminance, contrast, and structure between the two images. SSIM operates on local windows and is more sensitive to structural distortions than purely voxel-wise metrics, making it particularly relevant for assessing anatomical coherence.

Region-Level Metrics

The Dice similarity coefficient measures the overlap between the anatomical parcellations of the synthesised and ground-truth scans for each region of interest. Given a region R with predicted segmentation $\hat{S}_R$ and ground-truth segmentation S_R, the Dice coefficient is defined as:

$$\text{Dice}(S_R, \hat{S}_R) = \frac{2|S_R \cap \hat{S}_R|}{|S_R| + |\hat{S}_R|} \tag{7}$$

We compute Dice scores for eight brain regions relevant to Alzheimer's disease: hippocampus, parahippocampal gyrus, third ventricle, amygdala, entorhinal cortex, fusiform gyrus, inferior temporal gyrus, and middle temporal gyrus. These regions were selected on the basis of their known vulnerability to neurodegenerative atrophy and their established role as structural biomarkers in AD progression [].

In addition, we report the absolute volumetric difference (in voxels) and the relative volumetric difference (as a percentage of the ground-truth volume) for each region. This measures whether the synthesised scan preserves the volumetric properties of each structure, which is essential for downstream applications such as atrophy rate estimation and longitudinal biomarker tracking.

Deformation-Level Metrics

To assess the anatomical plausibility of the predicted deformation fields, we examine the Jacobian determinant of the SVF at each spatial location.

The Jacobian determinant $\det(J_{\hat{\phi}}(x))$ quantifies the local volume change induced by the deformation: values equal to 1 indicate volume preservation, values below 1 indicate contraction, and values above 1 indicate expansion. Negative values would indicate topology-violating folding, which is anatomically implausible. We report the mean and standard deviation of the Jacobian determinant across the brain volume for each test fold. In a cohort that includes subjects with MCI and AD, a mean Jacobian slightly below unity—reflecting mild net tissue atrophy—is expected and consistent with the known neurodegeneration patterns in these populations.

5. Results and Discussion

5.1. Evaluation of SynBADD Variants

We analyse the contribution of each module introduced in SynBADD through a series of controlled ablation experiments. The evaluation is organised along two axes: the design of the temporal transformer that operates in the latent space, and the spatial refinement stage that operates on the decoded SVF. All results are averaged over five-fold cross-validation.

5.1.1. Effect of Temporal Transformer Design

We compare three variants of the cross-attention mechanism used to aggregate temporal information from the observed latent codes. In the base variant, cross-attention is applied directly, using the target-aware query to pool over the encoded sequence. In the second variant, a learned sigmoid gate is introduced to scale per-token attention scores before pooling, enabling the model to suppress contributions from less informative timepoints. In the third variant, the order is reversed: gating is applied first to modulate the encoded tokens, after which cross-attention is used to aggregate the gated representations with respect to the target query.

Table 1 reports the results. Performance is broadly similar across all three variants, with PSNR values ranging from 33.39 to 33.49 and SSIM remaining stable around 0.9882–0.9883. A minor improvement in PSNR is observed when gating is applied after cross-attention, suggesting that post-hoc suppression of noisy signals may offer a slight advantage over pre-filtering. However, the base cross-attention variant achieves the highest SSIM overall.

The consistency across variants indicates that the transformer architecture is relatively robust to the specific form of the attention mechanism.

This is likely because the latent codes, having been extracted from a pre-trained VAE, already constitute a compact and structured representation in which the temporal signal is well-separated from noise. In the subsequent experiments, we adopt the base cross-attention variant for its simplicity and marginally stronger structural similarity.

Table 1: Comparison of our model with and without gating. Averaged over 5-fold cross-validation. ↑ indicates higher is better.

Metric ↑	Cross Attention	Cross Attention + Gating	Gating + Cross Attention
PSNR	33.417814	**33.4939**	33.392662
SSIM	**0.988284**	0.98826	0.988202

5.1.2. Effect of SVF Refinement

The SVF decoded from the predicted latent code provides a coarse estimate of the deformation field at the missing timepoint. To assess whether spatial refinement improves this estimate, we compare three configurations: (i) the full model with U-Net refinement and clinical conditioning via DAFT, (ii) refinement with U-Net alone, and (iii) no refinement module.

Table 2 summarises the results. Introducing the U-Net refinement module improves PSNR by 1.90% over the variant without any refinement, with a corresponding increase in SSIM from 0.9857 to 0.9876. Adding DAFT conditioning on top of the U-Net yields a further improvement, bringing PSNR to 33.42 and SSIM to 0.9883. The cumulative effect of the full refinement pipeline amounts to a PSNR gain of 3.16% relative to the no-refinement baseline.

These results confirm that the coarse SVF reconstructed from the latent code, while capturing the overall trajectory of deformation, lacks the spatial precision required to faithfully represent fine-grained anatomical structures. The refinement module operates at full resolution and adjusts local deformation patterns that the low-dimensional bottleneck cannot preserve. We also observed that the qualitative alignment with target anatomy improves noticeably with the full refinement pipeline, particularly in regions exhibiting complex cortical folding such as the temporal and parietal lobes. This improvement is not fully reflected in PSNR and SSIM, which are global metrics and may underweight localised gains in anatomically critical areas.

Table 2: Comparison of our model with and without SVF Refinement (UNET) and with and without clinical conditioning (DAFT). Averaged over 5-fold cross-validation. ↑ indicates higher is better.

Metric ↑	UNet + DAFT	Only UNet	No Unet+ No DAFT
PSNR	**33.417814**	33.008994	32.3926225
SSIM	**0.988284**	0.987614	0.9856825

5.1.3. Impact of Clinical Conditioning

In the full model, the U-Net refinement module is conditioned via DAFT using a 3-dimensional auxiliary vector composed of diagnostic group, subject age, and the target timepoint encoding. Concretely, the conditioning input is formed as $x_{\text{aux}} = [\text{group},\ \text{age},\ t_{\text{target}}]$, where group is encoded as CN = 0, MCI = 1, and AD = 2. This design allows the refinement network to modulate predicted deformations according to both subject-specific disease stage and the temporal horizon of the synthesis task. Age provides a strong prior on the rate of neurodegeneration, as it directly influences baseline brain morphology and the expected trajectory of volumetric decline. Diagnostic label encodes disease-stage information that can modulate both the magnitude and spatial distribution of predicted deformations. The target timepoint encoding enables the network to scale deformation magnitude to the requested longitudinal horizon.

From Table 2, DAFT conditioning provides an additional PSNR improvement of approximately 1.24% over the U-Net–only variant, with a corresponding marginal increase in SSIM. The quantitative gain from conditioning is moderate in global metrics, which is not unexpected given that age and diagnosis are scalar-valued signals being injected into a high-dimensional spatial field. To assess whether the benefit of conditioning is anatomically localised, Table 3 reports Dice overlap across eight regions of interest that are highly relevant to Alzheimer's disease progression.

The U-Net–only variant performs worst across all evaluated ROIs, indicating that spatial refinement without auxiliary conditioning can degrade regional boundary accuracy despite improving global image fidelity (PSNR/SSIM). In contrast, adding DAFT conditioning substantially recovers this loss and yields the best mean Dice across regions (0.9578), with the highest scores in 5 of 8 ROIs: entorhinal cortex, fusiform gyrus, inferior temporal gyrus, middle temporal gyrus, and third ventricle.

Table 3: Dice coefficient comparison across model variants for eight AD-relevant brain regions. ↑ indicates higher is better. Bold denotes best per region.

Region	No U-Net + No DAFT	Only U-Net	U-Net + DAFT
Hippocampus	**0.9721**	0.9560	0.9703
Amygdala	**0.9662**	0.9525	0.9659
Entorhinal Cortex	0.9223	0.8931	**0.9256**
Parahippocampal	**0.9592**	0.9385	0.9537
Fusiform Gyrus	0.9342	0.9079	**0.9378**
Inferior Temporal Gyrus	0.9364	0.9040	**0.9576**
Middle Temporal Gyrus	0.9447	0.9053	**0.9707**
Third Ventricle	0.9806	0.9714	**0.9811**
Mean	0.9519	0.9286	**0.9578**

The largest gains of U-Net+DAFT over the U-Net–only variant are observed in the inferior temporal gyrus (+0.054), middle temporal gyrus (+0.065), entorhinal cortex (+0.033), and fusiform gyrus (+0.030), with a smaller gain in the third ventricle (+0.010). This pattern is anatomically meaningful: Alzheimer's disease is known to affect the medial temporal lobe early—including the entorhinal cortex and hippocampal system—before extending into broader inferior and middle temporal neocortical regions, while ventricular enlargement is a well-established macroscopic structural marker of disease progression [1, 2]. The clustering of DAFT gains in these structures suggests that conditioning on disease group, age, and target timepoint helps the model generate anatomically more plausible deformations in regions that are especially sensitive to neurodegenerative change.

Interestingly, the no-refinement baseline remains marginally superior in the hippocampus, amygdala, and parahippocampal region. A plausible explanation is that the refinement network, when left unconditioned, can slightly perturb already-good coarse deformations around small, tightly folded medial temporal structures where Dice is sensitive to even minor boundary shifts. DAFT conditioning recovers much of this loss but does not fully exceed the baseline in those three ROIs. This is consistent with the global results: adding the U-Net alone improves PSNR and SSIM, but the ROI analysis reveals that this improvement is not uniformly distributed across structures. The refinement module helps globally but can be locally heavy-handed unless guided by subject-specific metadata.

Taken together, the ablation results confirm that each component of SynBADD contributes to overall performance. The temporal transformer provides a stable mechanism for aggregating longitudinal context, the refinement module recovers spatial detail lost during latent decoding, and clinical conditioning introduces subject-specific priors that concentrate improvements in disease-relevant anatomy. The Dice analysis complements the PSNR and SSIM results by showing that the benefit of DAFT is not merely global but is concentrated in anatomical regions implicated in Alzheimer's disease progression.

5.2. Comparison with State-of-the-Art Methods

We benchmark SynBADD against two recent methods for missing scan imputation in longitudinal brain MRI: NODER [], a neural ODE–based framework that learns subject-specific deformation trajectories through curve fitting in SVF space, and ReMiND [], a diffusion model that synthesises missing scans slice-by-slice in intensity space, conditioned on the immediate preceding and succeeding timepoints. We additionally compare against a simple Forward Difference baseline, in which the closest available preceding scan is used as a substitute for the missing target.

As each scan is of size $128 \times 128 \times 128$ and the brain occupies on average approximately 50% of the volume, we apply brain-masking to extract the brain region and compute all metrics exclusively on this masked area. This avoids inflation of similarity scores by background voxels and ensures a fair comparison across methods.

NODER and ReMiND impose different requirements on the available temporal context. ReMiND requires both the immediate preceding and immediate succeeding scans to be present, whereas NODER does not enforce such a constraint. Accordingly, the evaluation sets for the two baselines differ: the test set for ReMiND is a subset of that used for NODER. We evaluate SynBADD under both settings, using the same test set as the corresponding baseline in each case, to ensure a fair comparison.

5.2.1. Comparison based on Image Quality Metrics

Table 4 reports image-level metrics for SynBADD and NODER, averaged over five-fold cross-validation. SynBADD achieves a PSNR of 34.52 compared to 31.40 for NODER, representing an improvement of 9.94%. SSIM improves from 0.9766 to 0.9912, a gain of 1.50%. Table 5 presents the corresponding

comparison with ReMiND. Here the margin is substantially larger: SynBADD achieves a PSNR improvement of 38.18% and an SSIM improvement of 4.27% over ReMiND.

The magnitude of the improvement over ReMiND reflects a fundamental difference in generation strategy. ReMiND synthesises each 2D slice independently and stacks them to approximate a 3D volume, which introduces inter-slice inconsistencies and limits volumetric coherence. SynBADD, by contrast, operates natively in 3D deformation space, producing spatially coherent volumes by construction. The improvement over NODER, while smaller in relative terms, is nonetheless substantial and can be attributed to SynBADD's ability to leverage population-level learning and bidirectional temporal context, whereas NODER fits a separate trajectory per subject and per timepoint.

Table 4: State-of-the-art (SOTA) benchmarking of our model **SynBADD** against **NODER** on image quality metrics. Averaged over 5-fold cross-validation. ↑ indicates higher is better.

Metric ↑	NODER	SynBADD
PSNR	31.40	**34.52**
SSIM	0.9766	**0.9912**

Table 5: Benchmarking of **SynBADD** against **ReMiND** on image quality metrics. Results are averaged over 5-fold cross-validation. ↑ indicates higher is better.

Metric ↑	ReMiND	SynBADD
PSNR	24.28	**33.55**
SSIM	0.9511	**0.9917**

5.2.2. Comparison on Clinically Relevant metrics

Beyond global image quality, we evaluate anatomical fidelity by computing Dice coefficients between FreeSurfer parcellations of the synthesised and ground-truth scans across eight brain regions relevant to Alzheimer's disease progression: hippocampus, parahippocampal gyrus, third ventricle, amygdala, entorhinal cortex, fusiform gyrus, inferior temporal gyrus, and middle temporal gyrus.

Table 6 shows that SynBADD achieves Dice improvements over NODER ranging from +28.83% for the third ventricle to +82.55% for the middle temporal gyrus. The largest gains are observed in regions where NODER produces the weakest overlap with ground truth, such as the entorhinal cortex (0.5168 vs. 0.9209) and parahippocampal gyrus (0.5742 vs. 0.9612). These are precisely the regions most affected by early Alzheimer's pathology, where accurate imputation is clinically most relevant.

Table 7 presents the comparison with ReMiND. SynBADD achieves consistently high Dice scores across all regions, exceeding 0.96 in every structure. ReMiND, by contrast, produces Dice scores below 0.67 in most regions, with the entorhinal cortex showing the weakest agreement at 0.3742. The gap is particularly pronounced in smaller structures such as the amygdala and entorhinal cortex, where slice-wise synthesis in intensity space struggles to maintain anatomical correspondence.

Table 8 and Table 9 extend the comparison to the Forward Difference baseline. Table 8 shows that SynBADD achieves Dice improvements ranging from +56.30% for the hippocampus to +110.94% for the middle temporal gyrus. Table 9 reports absolute and relative volumetric differences with respect to ground truth. The Forward Difference approach yields percentage volume errors in the range of 7.23%–20.34%, whereas SynBADD reduces these to 1.22%–3.79% across all evaluated regions. This indicates that SynBADD not only reconstructs the correct spatial layout of anatomical structures but also preserves their volumetric properties with high fidelity—a property that is essential for downstream tasks such as atrophy rate estimation and biomarker extraction in longitudinal studies.

Figure 3 visualises the Dice comparisons across brain regions, illustrating the consistent advantage of SynBADD over both NODER and ReMiND across all evaluated structures.

Table 8 shows that **SynBADD** achieves substantial Dice improvements in key ROIs relevant for Alzheimer's disease over the **Forward** baseline, ranging from +56.30% for the hippocampus to +110.94% for the middle temporal gyrus.

Table 9 reports the absolute and relative volume differences with respect to ground truth for SynBADD and a simple **Forward Difference** imputation baseline, in which the closest previous timepoint is used to approximate the missing target. Across all evaluated brain regions, SynBADD consistently yields substantially lower volume errors than the Forward Difference approach, both in absolute terms and as a percentage of the true volume.

Table 6: Dice coefficient comparison between **SynBADD** and **NODER** across brain regions. ↑ indicates higher is better.

Region	NODER Dice ↑	SynBADD Dice ↑
Hippocampus	0.6898	**0.9562**
Parahippocampal	0.5742	**0.9612**
Third Ventricle	0.7647	**0.9852**
Amygdala	0.6371	**0.9705**
Entorhinal Cortex	0.5168	**0.9209**
Fusiform Gyrus	0.5726	**0.9679**
Inferior Temporal Gyrus	0.5512	**0.9781**
Middle Temporal Gyrus	0.5364	**0.9792**

Table 7: Dice coefficient comparison between **SynBADD** and **ReMiND** across brain regions. ↑ indicates higher is better.

Region	ReMiND Dice ↑	SynBADD Dice ↑
Hippocampus	0.6601	**0.9893**
Parahippocampal	0.5658	**0.9709**
Third Ventricle	0.7228	**0.9883**
Amygdala	0.6461	**0.9801**
Entorhinal Cortex	0.3742	**0.9633**
Fusiform Gyrus	0.5119	**0.9832**
Inferior Temporal Gyrus	0.4356	**0.9875**
Middle Temporal Gyrus	0.4851	**0.9952**

Percentage errors are reduced from $7.23\% - 20.34\%$ with Forward Difference to $1.22\% - 3.79\%$ with SynBADD, indicating markedly improved volumetric fidelity, particularly in regions known to undergo progressive atrophy in Alzheimer's disease.

5.2.3. Classification of Comparative Methods

To contextualise the performance differences reported above, we characterise each method along several design dimensions that influence imputation quality. These are summarised in Table **??**.

ReMiND operates in the intensity space, generating missing scans by synthesising 2D slices conditioned on the immediately adjacent timepoints. It

Table 8: Comparison of Dice scores between **SynBADD** and **Forward** across brain regions. Values represent average Dice coefficients. ↑ indicates higher is better.

Region	Forward Dice ↑	SynBADD Dice ↑	Dice Diff
Hippocampus	0.595	**0.930**	+0.336
Parahippocampal	0.437	**0.903**	+0.466
Third Ventricle	0.598	**0.973**	+0.376
Amygdala	0.582	**0.916**	+0.334
Entorhinal Cortex	0.471	**0.855**	+0.384
Fusiform Gyrus	0.432	**0.864**	+0.432
Inferior Temporal Gyrus	0.449	**0.849**	+0.400
Middle Temporal Gyrus	0.403	**0.850**	+0.447

Table 9: Benchmarking of **SynBADD** against **Forward** differences on volume difference compared to the ground-truth brain regions. Values represent absolute (voxels) and percentage differences.

Region	Forward Diff	Forward Diff (%)	SynBADD Diff	SynBADD Diff (%)
Hippocampus	561.270	10.839	108.038	1.818
Parahippocampal	300.140	10.918	78.603	2.507
Third Ventricle	226.921	20.340	17.473	1.222
Amygdala	289.803	16.109	50.108	2.448
Entorhinal Cortex	288.660	12.978	90.032	3.794
Fusiform Gyrus	860.089	8.683	252.063	2.278
Inferior Temporal Gyrus	1038.476	7.305	328.559	2.094
Middle Temporal Gyrus	1193.117	7.229	259.860	1.526

does not enforce volumetric consistency across slices, nor does it impose explicit anatomical constraints on the generated outputs. Its temporal reasoning is limited to pairwise interpolation between two flanking scans, without access to the broader longitudinal trajectory.

NODER models deformations using stationary velocity fields and learns a continuous trajectory in SVF space via neural ordinary differential equations. However, it requires fitting a separate model for each subject and each target timepoint, which limits scalability and introduces susceptibility to overfitting, particularly when the number of available scans per subject is small. Its supervision is based on image regression, which may lead to oversmoothing of predicted deformations.

SynBADD shares with NODER the principle of operating in deformation space, but differs in several respects. It trains a single population-level model

that generalises across subjects and diagnostic groups, rather than requiring per-subject optimisation. It reasons bidirectionally over the full available longitudinal context, rather than relying on pairwise or unidirectional conditioning. The refinement module, conditioned on clinical metadata, provides subject-specific adaptation within the population-level framework. And the synthesis is performed natively in 3D, producing volumetrically coherent outputs by construction.

These design choices collectively account for the observed performance gap. Operating in deformation space preserves anatomical topology, bidirectional attention captures richer temporal context, and population-level training enables robust generalisation even under limited longitudinal sampling.

5.2.4. Quantitative Comparison on Full Evaluation Cohort

To assess the consistency of the reported improvements, we present per-fold results across all five cross-validation splits. Table **??** reports PSNR and SSIM for SynBADD and NODER in each fold. SynBADD outperforms NODER in every fold, with per-fold PSNR improvements ranging from 3.07 to 4.13 dB and SSIM improvements ranging from 0.0217 to 0.0268. The variance across folds is small for both methods, confirming that the observed gains are stable and not driven by favourable splits.

Table **??** extends this analysis to Dice scores for the hippocampus, parahippocampal gyrus, and third ventricle. SynBADD maintains Dice scores above 0.85 in every fold for all three regions, whereas NODER exhibits greater variability and consistently lower overlap, particularly for the parahippocampal gyrus.

We additionally report the Jacobian determinant statistics of the predicted deformation fields in Table **??**. Across all five folds, SynBADD produces mean Jacobian values in the range of 0.9232–0.9309, with standard deviations of approximately 0.04. A mean Jacobian slightly below unity is consistent with mild tissue contraction, reflecting the predominant atrophy pattern expected in a cohort that includes subjects with MCI and AD. Importantly, no negative Jacobian values were observed, confirming that all predicted deformations are diffeomorphic and topologically valid. This provides additional evidence that SynBADD generates anatomically plausible transformations, not merely transformations that produce visually similar images.

5.2.5. Subgroup Analysis Across Clinical and Temporal Variables

5.3. Robustness to Incomplete Temporal Context

In clinical practice, subjects rarely present with complete longitudinal imaging sequences. Missed visits, late enrolment, and early dropout frequently result in irregular and sparse temporal sampling. A practical imputation model must therefore degrade gracefully when fewer observations are available, and must handle asymmetric temporal context—where, for instance, more past scans than future scans are accessible, or vice versa. In this section, we evaluate SynBADD's sensitivity to these conditions.

5.3.1. Effect of Reduced Input Timepoints

5.3.2. Effect of Asymmetric Context Availability

5.4. Qualitative Analysis of Synthesized Outputs

Quantitative metrics capture global image fidelity but may not reveal localised anatomical artifacts or subtle structural inconsistencies that are relevant for clinical interpretation. In this section, we complement the numerical evaluation with qualitative comparisons of synthesised scans across methods, examining both the final warped images and the underlying predicted deformation fields.

5.4.1. Visual Comparison of Predicted Scans

Figures **??** and **??** present consecutive axial slices (slices 62–66) of synthesised scans for two representative subjects, comparing SynBADD, NODER, and ReMiND against the ground-truth target.

ReMiND, which generates each 2D slice independently before stacking them into a volume, exhibits noticeable inter-slice inconsistencies. Intensity discontinuities and hole-like artifacts are visible between adjacent slices, most prominently between slices 63 and 64, where both signal quality and anatomical coherence degrade. These spatial discontinuities compromise the volumetric integrity of the reconstructed scan and would limit its reliability in any downstream analysis that depends on 3D spatial relationships, such as segmentation or morphometric measurement.

NODER, as a deformation-based method, maintains volumetric continuity across slices. However, it tends to overestimate local deformations in certain regions, producing protrusion-like artifacts where tissue boundaries are displaced beyond their ground-truth positions. This is consistent with

the observation that NODER fits subject-specific trajectories through regression, which can amplify small estimation errors into localised geometric distortions.

SynBADD preserves smooth anatomical transitions across all consecutive slices, with no visible discontinuities or exaggerated deformations. The cortical folds, ventricular boundaries, and subcortical structures are consistently rendered across the slice sequence, reflecting the inherent spatial coherence of its 3D deformation-based synthesis.

Figure **??** presents voxel-wise difference maps between the synthesised and ground-truth scans, displayed as signed error fields in which blue indicates negative intensity differences, red indicates positive differences, and white corresponds to zero error. SynBADD achieves substantially lower error magnitudes across the brain volume compared to both baselines. ReMiND produces widespread errors, particularly in cortical and ventricular regions, reflecting the cumulative effect of slice-wise synthesis artifacts across the volume. NODER shows more localised but pronounced errors along tissue boundaries, consistent with its tendency to overestimate deformation magnitudes. The error distribution of SynBADD is both smaller in magnitude and more spatially uniform, with residual errors concentrated primarily in regions where the ground-truth deformation itself is largest—an expected pattern for a method that models deformation rather than absolute appearance.

Figure **??** provides a detailed view of SynBADD's capacity to capture subtle, localised anatomical changes. In this visualisation, the baseline scan is warped to the target timepoint using the predicted SVF, and the resulting image is compared against both the ground truth and the original baseline. Regions where fine-grained atrophy has been accurately captured by the warping are highlighted, alongside regions where the warped scan diverges from the undeformed baseline. The comparison confirms that SynBADD is sensitive to small but clinically meaningful changes—such as hippocampal volume reduction and ventricular expansion—that are characteristic of early neurodegenerative progression. This sensitivity is a direct consequence of operating in the deformation space, where the model's entire generative capacity is directed toward estimating the transformation rather than reconstructing the full intensity volume from scratch.

5.4.2. Visualization of Predicted Deformation Fields

To provide further insight into what the model has learned, we visualise the predicted stationary velocity fields directly. Figure **??** displays the SVF

for a representative subject, rendered as an RGB-mapped velocity field overlaid on axial, coronal, and sagittal views, where each colour channel encodes the velocity component along one spatial axis and the intensity reflects the local magnitude of the deformation.

The predicted deformations are spatially concentrated in regions known to be vulnerable to neurodegeneration in Alzheimer's disease: the medial temporal lobe, hippocampal formation, and periventricular areas. Deformation magnitudes are minimal in stable structures such as the brainstem and cerebellum, which is consistent with the known spatial distribution of AD-related atrophy. This pattern confirms that the model has not learned arbitrary spatial warps, but rather biologically interpretable transformations that reflect the expected progression of neurodegeneration.

We additionally examine the Jacobian determinant of the predicted deformation field, which quantifies local volume change at each spatial location. Values below unity indicate tissue contraction, while values above unity indicate expansion. Figure **??** shows the Jacobian determinant map for a representative case, overlaid on the corresponding anatomical slice. Regions of mild contraction (Jacobian values slightly below 1.0) are localised to the hippocampus, parahippocampal gyrus, and temporal cortex, while the ventricular system shows slight expansion—a pattern that is characteristic of progressive brain atrophy and is consistent with the literature on volumetric changes in AD. The absence of negative Jacobian values across the field confirms that the predicted deformation is diffeomorphic, preserving the topology of the underlying anatomy.

These qualitative analyses reinforce the quantitative findings and provide interpretable evidence that SynBADD produces not only metrically accurate but also anatomically meaningful and clinically plausible imputations.

6. Conclusion

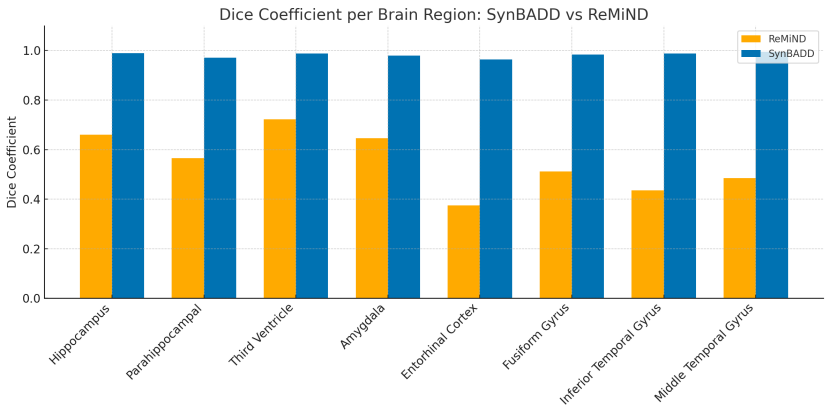
Appendix A. Example Appendix Section

Appendix text.

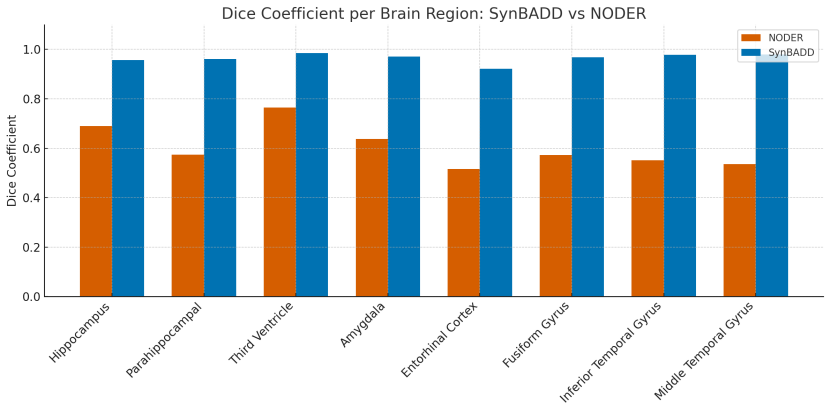
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(a) Dice coefficient per brain region: SynBADD vs ReMiND.



(b) Dice coefficient per brain region: SynBADD vs NODER.

Figure 3: Dice coefficient comparisons across brain regions between SynBADD, NODER, and ReMiND.